

Coordinate-free Mathematics

Paul Gustafson

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The first section develops algebraic constructions independent of a basis; the second develops analytic constructions from norms, operators, and positivity; the third develops geometric constructions independent of coordinates.

1 Algebra

Linear algebra studies linear maps between vector spaces. A **vector space** V over a field k is a set equipped with vector addition and scalar multiplication operations satisfying straightforward compatibility conditions. A **linear map** $f : V \rightarrow U$ between k -vector spaces is a map of underlying sets such that $f(\alpha v + \beta w) = \alpha f(v) + \beta f(w)$ for all $\alpha, \beta \in k$ and $v, w \in V$. Examples include differential operators, group representations, coefficient matrices of linear systems of equations, and local approximations of globally nonlinear functions [1, 2].

The **free k -vector space on a set** X , denoted kX , is the set of finitely supported set-theoretical functions $X \rightarrow k$ with pointwise addition and scalar multiplication. The **direct sum** vector space $V \oplus W$ is the set $V \times W$ with pointwise addition and scalar multiplication defined by $\alpha(v \oplus w) = \alpha v \oplus \alpha w$. The **quotient** of V by a subspace W , denoted V/W , is the set of equivalence classes of V modulo the relation $x \sim y$ iff $x - y \in W$. The **tensor product** $V \otimes W$ is the vector space $k(V \times W)/N$ where N is the subspace generated by the obvious linearity relations on V and W .

Exercise 1 An **isomorphism** of vector spaces $V \cong W$ consists of a pair of linear maps $f : V \rightarrow W$ and $g : W \rightarrow V$ such that $f \circ g = 1_W$ and $g \circ f = 1_V$. Prove the following:

- $k(X \sqcup Y) \cong kX \oplus kY$
- $k(X \times Y) \cong kX \otimes kY$
- The first isomorphism theorem for vector spaces (and second and third if you want).
- Find universal properties determining up to canonical isomorphism the free k -vector space, direct sum, quotient, tensor product, and dual space (defined below), respectively.

The **dual space** V^* is the space of linear maps from a vector space V to its ground field k . The evaluation map $\text{ev}_V : V^* \otimes V \rightarrow k$ is the linear extension of the evaluation morphism $f \otimes x \mapsto f(x)$. We say that V is **finite dimensional** if there is a co-evaluation morphism $\text{coev}_V : k \rightarrow V \otimes V^*$ satisfying the snake equations $(1_V \otimes \text{ev}_V) \circ (\text{coev}_V \otimes 1_V) = 1_V$ and $(\text{ev}_V \otimes 1_{V^*}) \circ (1_{V^*} \otimes \text{coev}_V) = 1_{V^*}$. We define the **dimension** $\dim(V) = \text{tr}(1_V)$, where $\text{tr}(f) : \text{End}(V) \rightarrow k$ is given by $f \mapsto (\text{ev}_V \circ s_{V,V^*} \circ (f \otimes 1_{V^*}) \circ \text{coev}_V)(1)$ where $s_{X,Y} : X \otimes Y \rightarrow Y \otimes X$ is the symmetric braiding $x \otimes y \mapsto y \otimes x$ [5].

Exercise 2 Prove the following for finite-dimensional vector spaces V, W and linear maps $f, g : V \rightarrow V$.

- $\text{tr}(fg) = \text{tr}(gf)$
- $\text{tr}(f \oplus g) = \text{tr}(f) + \text{tr}(g)$
- $\text{tr}(f \otimes g) = \text{tr}(f) \text{tr}(g)$
- $\dim(V) \in \mathbf{N}$
- $V \cong W$ iff $\dim(V) = \dim(W)$

We can solve for the eigenvalues of a linear operator on a finite-dimensional vector space via the determinant. Define the tensor algebra and **exterior algebra** of V by [4]

$$T(V) = \bigoplus_{k \geq 0} V^{\otimes k}, \quad \Lambda V = T(V) / \langle v \otimes v : v \in V \rangle.$$

The space $\Lambda^k V$ is the degree- k component of ΛV .

Exercise 3 Show that the highest graded component of ΛV is one-dimensional.

The **determinant** $\det(f)$ is the coefficient of the linear map on the highest-graded component of ΛV given by $v_1 \wedge \cdots \wedge v_{\dim(V)} \mapsto f(v_1) \wedge \cdots \wedge f(v_{\dim(V)})$.

For an n -dimensional real vector space V , its **determinant line** is $\det(V^*) = \Lambda^n V^*$. An **orientation** is a choice of one of the two connected components of $\det(V^*) \setminus \{0\}$; the elements of the chosen component are positive.

Exercise 4 Prove the following:

- $\det(f \oplus g) = \det(f) \det(g)$
- If $V = \mathbb{R}^n$, then $\det(f)$ is the signed volume of $f(I^n)$ where $I = [0, 1]$.
- Cramer's rule

For $f \in \text{End}(V)$, the **characteristic polynomial** is the polynomial $\lambda \mapsto \det(f - \lambda 1_V)$.

A scalar $\lambda \in k$ is an **eigenvalue** of f if $\ker(f - \lambda 1_V) \neq 0$. This kernel is its **eigenspace**; its dimension is the geometric multiplicity of λ .

Exercise 5 Assume that the characteristic polynomial of f splits over k . Prove the following for a linear operator f on a finite-dimensional vector space V :

- Every eigenvalue of f is a root of its characteristic polynomial.
- The geometric multiplicity of each eigenvalue is at most equal to its **algebraic multiplicity** (i.e. multiplicity as a root of the characteristic polynomial)
- $\text{tr}(f) = \sum_i \lambda_i$ and $\det(f) = \prod_i \lambda_i$, where each eigenvalue λ_i is repeated according to its algebraic multiplicity
- The operator f satisfies its own characteristic polynomial (replacing λ with f)
- The existence of the Jordan normal form for a linear operator on a finite-dimensional complex vector space

A $\mathbb{C}$ -linear monoidal category generalizes finite-dimensional complex vector spaces: it has objects, complex vector spaces $\text{Hom}_{\mathcal{C}}(x, y)$ of morphisms, identities and bilinear associative composition, together with a tensor product $\otimes$ that is bilinear on morphisms and is associative with tensor unit 1, up to coherent natural isomorphism. Direct sums and duals generalize the corresponding vector-space constructions.

Let $\mathcal{C}$ be such a category with finite direct sums. An object x is **simple** if $\text{End}(x) \cong \mathbb{C}$, and $\mathcal{C}$ is **semisimple** if every object is a finite direct sum of simple objects. Assume that 1 is simple, that there are finitely many isomorphism classes of simple objects, and that $\mathcal{C}$ is **rigid**: every x has a dual x^* with evaluation and coevaluation maps satisfying the snake identities [37].

The definitions of duality, trace, and dimension above extend verbatim to $\mathcal{C}$. Write $d_x = \text{tr}(1_x)$. Then

$$d_{x \oplus y} = d_x + d_y, \quad d_{x \otimes y} = d_x d_y.$$

Assume that the categorical dimensions of nonzero objects are positive real numbers.

For $x, y, z \in \mathcal{C}$, write

$$F^{xyz} : (x \otimes y) \otimes z \longrightarrow x \otimes (y \otimes z)$$

for the associator. A **braiding** is a family of isomorphisms

$$c_{x,y} : x \otimes y \longrightarrow y \otimes x.$$

For $f : x \rightarrow x'$ and $g : y \rightarrow y'$, it satisfies $(g \otimes f)c_{x,y} = c_{x',y'}(f \otimes g)$. The associator satisfies the pentagon identity, and the associator and braiding satisfy the hexagon identities. Suppressing associators, the latter are

$$\begin{aligned} c_{x,y \otimes z} &= (1_y \otimes c_{x,z})(c_{x,y} \otimes 1_z), \\ c_{x \otimes y,z} &= (c_{x,z} \otimes 1_y)(1_x \otimes c_{y,z}). \end{aligned}$$

The **monodromy** of x and y is the double braiding

$$M_{x,y} = c_{y,x}c_{x,y} : x \otimes y \longrightarrow x \otimes y.$$

Thus $c_{x,y}$ is one crossing, while $M_{x,y}$ is a full braid. If $x \otimes y$ is simple, identify $M_{x,y}$ with its scalar in $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. Decompositions of tensor products of simple objects are **fusion rules**, and the spaces $\text{Hom}_{\mathcal{C}}(z, x \otimes y)$ are **fusion spaces**.

An object x is **invertible** if $x \otimes x^* \cong 1$. Assume first that every simple object is invertible and that their isomorphism classes form the cyclic group

$$\{1, a, a^2, \dots, a^{m-1}\}, \quad a^r \otimes a^s \cong a^{r+s \pmod{m}}.$$

Write $U(1) = \{z \in \mathbb{C} : |z| = 1\}$ and $M_{r,s} = M_{a^r, a^s} \in U(1)$.

Exercise 6 (Abelian braiding phase)

Use the hexagon identities to prove that

$$M_{r+s,t} = M_{r,t}M_{s,t}, \quad M_{r,s+t} = M_{r,s}M_{r,t}.$$

Deduce that there is an integer k modulo m such that

$$M_{r,s} = \exp\left(\frac{2\pi i k r s}{m}\right).$$

For $k = 1$, prove that the monodromy of a about N copies of a is multiplication by

$$\exp\left(\frac{2\pi i N}{m}\right).$$

In particular, for $m = 3$, increasing N by one changes the phase by $2\pi/3$.

The phase $2\pi/3$ is the value observed by interferometry in the $\nu = 1/3$ fractional quantum Hall state [39].

Now suppose that $\mathcal{C}$ has two simple objects $1, \tau$ satisfying the fusion rule

$$\tau \otimes \tau \cong 1 \oplus \tau.$$

Exercise 7 (Fibonacci categorical dimension)

Let $d_\tau = \text{tr}(1_\tau)$. Prove that [37]

$$d_\tau^2 = 1 + d_\tau$$

and, using positivity, that

$$d_\tau = \frac{1 + \sqrt{5}}{2}.$$

Let

$$V_n = \text{Hom}_{\mathcal{C}}(1, \tau^{\otimes n}).$$

Show that the dimensions of these fusion spaces satisfy the Fibonacci recurrence and determine their asymptotic growth rate.

Let

$$V = \text{Hom}_{\mathcal{C}}(\tau, \tau^{\otimes 3}).$$

This fusion space is two-dimensional, with basis indexed by the intermediate objects $1, \tau$ in $(\tau \otimes \tau) \otimes \tau$. Set $\varphi = (1 + \sqrt{5})/2$ and choose [38]

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad R = \begin{pmatrix} e^{-4\pi i/5} & 0 \\ 0 & e^{3\pi i/5} \end{pmatrix}.$$

Define $\sigma_1 = R$ and $\sigma_2 = F^{-1}RF$. Let

$$B_3 = \langle b_1, b_2 \mid b_1 b_2 b_1 = b_2 b_1 b_2 \rangle.$$

Exercise 8 (Non-Abelian braid representation)

Working exactly, verify that F and R are unitary and prove that

$$\sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2.$$

Prove also that $\sigma_1 \sigma_2 \neq \sigma_2 \sigma_1$. Deduce that $b_i \mapsto \sigma_i$ defines a non-Abelian representation

$$B_3 \longrightarrow U(V).$$

Such Fibonacci fusion and braiding data have been implemented on superconducting quantum processors [41].¹

2 Analysis

2.1 Norms, operator exponentials, and linear evolution

A **norm** on a complex vector space H is a function $\|\cdot\| : H \rightarrow [0, \infty)$ that is definite, homogeneous, and satisfies the triangle inequality. A normed space is a **Banach space** if every Cauchy sequence converges. A **Hilbert space** is a complex vector space with a Hermitian inner product $\langle \cdot, \cdot \rangle$ whose norm $\|x\| = \sqrt{\langle x, x \rangle}$ is complete. The Cauchy–Schwarz inequality

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

implies the triangle inequality. A linear map $A : H \rightarrow K$ is **bounded** if

$$\|A\| = \sup_{\|x\|=1} \|Ax\| < \infty.$$

Write $\mathcal{B}(H, K)$ for the Banach space of bounded operators and $\mathcal{B}(H) = \mathcal{B}(H, H)$.

For $A \in \mathcal{B}(H)$, the series

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}$$

converges in operator norm because $\|A^n\| \leq \|A\|^n$. Only this elementary consequence of power-series convergence is needed below.

¹This is an engineered quantum simulation of Fibonacci topological order, not evidence that naturally occurring Fibonacci quasiparticles have been conclusively observed. A related trapped-ion realization of non-Abelian topological order is reported in [40].

Exercise 9 (Operator exponentials and linear evolution)

Prove that e^{tA} is differentiable in operator norm and that

$$\frac{d}{dt}e^{tA} = Ae^{tA} = e^{tA}A.$$

Deduce that the unique solution of

$$\dot{x}(t) = Ax(t), \quad x(0) = x_0,$$

is $x(t) = e^{tA}x_0$. In finite dimensions use the Jordan decomposition

$$A = S + N, \quad [S, N] = 0,$$

with S semisimple and N nilpotent, to prove

$$e^{tA} = e^{tS}e^{tN}.$$

Show that on the generalized λ -eigenspace every component of a solution is a linear combination of terms $t^j e^{\lambda t}$.

2.2 Hilbert-space operator theory

For $A \in \mathcal{B}(H, K)$ there is a unique **adjoint** $A^* \in \mathcal{B}(K, H)$ satisfying

$$\langle Ax, y \rangle = \langle x, A^*y \rangle.$$

An operator $A \in \mathcal{B}(H)$ is **self-adjoint** if $A = A^*$, **positive** if $\langle x, Ax \rangle \geq 0$ for every x , and **unitary** if $A^*A = AA^* = 1$. In finite dimensions, positivity is equivalent to $A = B^*B$ for some B .

Exercise 10 (Spectral theorem and singular-value decomposition)

Let H be a finite-dimensional complex Hilbert space. Prove that every self-adjoint $A \in \text{End}(H)$ has an orthogonal spectral decomposition

$$A = \sum_{\lambda} \lambda p_{\lambda}, \quad p_{\lambda} p_{\mu} = \delta_{\lambda\mu} p_{\lambda}, \quad \sum_{\lambda} p_{\lambda} = 1.$$

Apply the theorem to T^*T to prove that every map $T : H \rightarrow K$ has a singular-value decomposition

$$T = U\Sigma V^*.$$

Prove that the diagonal entries of Σ are the square roots of the nonzero eigenvalues of T^*T , counted with multiplicity.

2.3 Quantum states and uncertainty

For a unit vector $\psi \in H$ and a self-adjoint observable $A \in \mathcal{B}(H)$, define its **expectation** and **variance** by

$$\langle A \rangle_{\psi} = \langle \psi, A\psi \rangle, \quad \text{Var}_{\psi}(A) = \|(A - \langle A \rangle_{\psi} 1)\psi\|^2.$$

Exercise 11 (Robertson uncertainty inequality)

For self-adjoint $A, B \in \mathcal{B}(H)$, apply Cauchy-Schwarz to the two centered vectors and take the imaginary part of their inner product to prove [25]

$$\mathrm{Var}_\psi(A) \mathrm{Var}_\psi(B) \geq \frac{1}{4} |\langle \psi, [A, B] \psi \rangle|^2, \quad [A, B] = AB - BA.$$

Determine the equality condition.

If H is self-adjoint, then $U_t = e^{-itH}$ is unitary and describes linear quantum evolution.

Exercise 12 (Conservation from commutators)

Let A, H be self-adjoint. Prove that [17]

$$\frac{d}{dt} \langle U_t \psi, A U_t \psi \rangle = -i \langle U_t \psi, [H, A] U_t \psi \rangle.$$

Deduce that $[H, A] = 0$ if and only if the expectation of A is constant under this evolution for every initial vector.

Exercise 13 (No-cloning theorem)

Let $b \in H$ be a unit vector and suppose that a unitary $C : H \otimes H \rightarrow H \otimes H$ satisfies $C(v \otimes b) = v \otimes v$ and $C(w \otimes b) = w \otimes w$ for unit vectors v, w . Prove that [24]

$$\langle v, w \rangle = \langle v, w \rangle^2.$$

Deduce that no unitary map clones every unit vector.

2.4 Schmidt decomposition and entanglement

For finite-dimensional Hilbert spaces H_A, H_B and $\psi \in H_A \otimes H_B$, define

$$T_\psi : H_A^* \rightarrow H_B, \quad T_\psi(\phi) = (\phi \otimes 1)(\psi).$$

The singular-value decomposition of T_ψ gives orthonormal families (a_i) and (b_i) and numbers $s_i > 0$ such that

$$\psi = \sum_{i=1}^r s_i a_i \otimes b_i.$$

This is the **Schmidt decomposition**; r is the Schmidt rank. For a unit vector ψ , the reduced density operators are

$$\rho_A = \mathrm{Tr}_{H_B}(|\psi\rangle\langle\psi|), \quad \rho_B = \mathrm{Tr}_{H_A}(|\psi\rangle\langle\psi|).$$

Exercise 14 (Schmidt decomposition and entanglement)

Prove that the nonzero eigenvalues of ρ_A and ρ_B are the numbers s_i^2 , with the same multiplicities. Prove that ψ is separable, meaning $\psi = a \otimes b$, exactly when its Schmidt rank is one. Deduce that the entanglement entropy is [17]

$$S(\psi) = -\mathrm{Tr}(\rho_A \log \rho_A) = -\sum_i s_i^2 \log s_i^2,$$

with the convention $0 \log 0 = 0$.

2.5 CHSH, Gisin, and Tsirelson

Let A_0, A_1 and B_0, B_1 be self-adjoint operators with eigenvalues in $\{-1, 1\}$. Their **CHSH operator** is

$$\mathcal{C} = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

Every classical local-hidden-variable assignment satisfies $|\langle \mathcal{C} \rangle| \leq 2$.

Exercise 15 (Gisin's theorem for two qubits)

For the two-qubit state in Schmidt form

$$\psi_\theta = \cos \theta |00\rangle + \sin \theta |11\rangle, \quad 0 \leq \theta \leq \frac{\pi}{4},$$

optimize over projective binary observables and prove that the maximal CHSH value is [18]

$$2\sqrt{1 + \sin^2(2\theta)}.$$

Conclude that every entangled pure two-qubit state violates a CHSH inequality. For $\theta = \pi/4$, recover the value $2\sqrt{2}$.

Tsirelson's representation theorem identifies every finite bipartite binary quantum correlation matrix with a Gram-type matrix [19]:

$$c_{ij} = \langle x_i, y_j \rangle$$

for unit vectors x_i, y_j in a real Hilbert space; conversely, every such finite family has a quantum realization. For a real matrix $M = (M_{ij})$, define

$$\beta_C(M) = \max_{\varepsilon_i, \delta_j \in \{-1, 1\}} \left| \sum_{i,j} M_{ij} \varepsilon_i \delta_j \right|$$

and

$$\beta_Q(M) = \sup_{\|x_i\|=\|y_j\|=1} \left| \sum_{i,j} M_{ij} \langle x_i, y_j \rangle \right|.$$

2.6 Grothendieck inequality

The real Grothendieck inequality states that there is a universal constant $K_G^{\mathbb{R}}$ such that [20, 21]

$$\beta_Q(M) \leq K_G^{\mathbb{R}} \beta_C(M)$$

for every real matrix M . Thus quantum mechanics gives at most a universal advantage for bipartite binary correlation Bell inequalities. CHSH has $\beta_C = 2$ and $\beta_Q = 2\sqrt{2}$, so

$$\sqrt{2} \leq K_G^{\mathbb{R}}.$$

Exercise 16 (CHSH as a Grothendieck ratio)

For

$$M_{\text{CHSH}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

compute $\beta_{\mathbb{C}}(M_{\text{CHSH}})$ directly. Choose four unit vectors in $\mathbb{R}^2$ that attain

$$\beta_{\mathbb{Q}}(M_{\text{CHSH}}) = 2\sqrt{2}$$

and verify the ratio $\sqrt{2}$.

2.7 C^* -algebras and positive maps

These finite-dimensional examples motivate a general language for normed algebras, positivity, and states. A **Banach algebra** is a Banach space A with an associative bilinear multiplication satisfying $\|ab\| \leq \|a\|\|b\|$.

A **unital C^* -algebra** is a complex unital Banach algebra A with an involution $a \mapsto a^*$ satisfying

$$\|a^*a\| = \|a\|^2.$$

Its positive cone is $A_+ = \{a^*a : a \in A\}$. A **state** is a linear map $\varphi : A \rightarrow \mathbb{C}$ such that [9]

$$\varphi(a) \geq 0 \quad (a \in A_+), \quad \varphi(1) = 1.$$

Exercise 17 Let φ be a positive linear functional on a unital C^* -algebra A . Prove that

$$|\varphi(b^*a)|^2 \leq \varphi(a^*a)\varphi(b^*b).$$

Deduce that $\|\varphi\| = \varphi(1)$.

For a state φ , define

$$\langle a, b \rangle_{\varphi} = \varphi(b^*a), \quad N_{\varphi} = \{a \in A : \varphi(a^*a) = 0\},$$

and let H_{φ} be the Hilbert-space completion of A/N_{φ} . For a Hilbert space H , write $\mathcal{B}(H)$ for the C^* -algebra of bounded operators on H . A vector Ω is **cyclic** for a representation π if the span of $\pi(A)\Omega$ is dense.

Exercise 18 (GNS construction)

Let φ be a state on a unital C^* -algebra A . Prove that left multiplication induces a unital $*$ -representation

$$\pi_{\varphi} : A \longrightarrow \mathcal{B}(H_{\varphi})$$

with a cyclic vector Ω_{φ} satisfying

$$\varphi(a) = \langle \pi_{\varphi}(a)\Omega_{\varphi}, \Omega_{\varphi} \rangle_{\varphi}.$$

Prove that $(H_{\varphi}, \pi_{\varphi}, \Omega_{\varphi})$ is unique up to unitary equivalence.

For a linear map $\Phi : A \rightarrow B$ between unital C^* -algebras, its n -th **amplification** is

$$\Phi_n : M_n(A) \longrightarrow M_n(B), \quad [a_{ij}] \longmapsto [\Phi(a_{ij})].$$

The map Φ is **completely positive** if every Φ_n is positive. Its **completely bounded norm** is

$$\|\Phi\|_{\text{cb}} = \sup_{n \geq 1} \|\Phi_n\|.$$

Exercise 19 For $n \geq 2$, let

$$T : M_n(\mathbb{C}) \longrightarrow M_n(\mathbb{C}), \quad T(a) = a^\top.$$

Prove that T is positive but not completely positive, and prove that

$$\|T\|_{\text{cb}} = n.$$

For the standard matrix units $E_{ij} \in M_n(\mathbb{C})$ and a linear map $\Phi : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$, define the **Choi matrix**

$$C_\Phi = \sum_{i,j=1}^n E_{ij} \otimes \Phi(E_{ij}).$$

Exercise 20 (Choi theorem)

Let $\Phi : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ be linear. Prove that the following are equivalent [10]:

- Φ is completely positive.
- $C_\Phi \geq 0$.
- There are linear maps $V_r : \mathbb{C}^m \rightarrow \mathbb{C}^n$ such that

$$\Phi(a) = \sum_r V_r^* a V_r.$$

Exercise 21 (Stinespring dilation)

Let $\Phi : A \rightarrow \mathcal{B}(H)$ be completely positive. Prove that there are a Hilbert space K , a $*$ -representation $\pi : A \rightarrow \mathcal{B}(K)$, and a bounded operator $V : H \rightarrow K$ such that [11]

$$\Phi(a) = V^* \pi(a) V.$$

Prove that V is an isometry when Φ is unital. Call the dilation minimal if the span of $\pi(A)VH$ is dense in K , and prove that a minimal dilation is unique up to unitary equivalence.

For a unital completely positive map $\Phi : A \rightarrow B$, define its **multiplicative domain** by

$$\text{MD}(\Phi) = \left\{ a \in A : \begin{array}{l} \Phi(a^*a) = \Phi(a)^* \Phi(a), \\ \Phi(aa^*) = \Phi(a) \Phi(a)^* \end{array} \right\}.$$

Exercise 22 (Multiplicative domain)

Let $\Phi : A \rightarrow B$ be unital and completely positive. Prove the Kadison–Schwarz inequality

$$\Phi(a)^* \Phi(a) \leq \Phi(a^* a).$$

Prove that $\text{MD}(\Phi)$ is a C^* -subalgebra of A and that, for $a \in \text{MD}(\Phi)$ and $b \in A$,

$$\Phi(ab) = \Phi(a)\Phi(b), \quad \Phi(ba) = \Phi(b)\Phi(a).$$

2.8 Completely bounded maps and operator spaces

An **operator space** is a closed linear subspace $E \subseteq \mathcal{B}(H)$, equipped with the matrix norms inherited from

$$M_d(E) \subseteq \mathcal{B}(H^{\oplus d}).$$

An abstract family of matrix norms comes from an operator space exactly when it satisfies Ruan’s axioms. If $T : E \rightarrow F$ is linear, its amplification is

$$T_d : M_d(E) \rightarrow M_d(F), \quad [x_{ij}] \mapsto [T(x_{ij})],$$

and T is **completely bounded** when $\|T\|_{\text{cb}} = \sup_d \|T_d\| < \infty$.

For a bilinear form $u : E \times F \rightarrow \mathbb{C}$, define its matrix amplification by

$$u_d([x_{ij}], [y_{kl}]) = [u(x_{ij}, y_{kl})]_{(i,k),(j,l)} \in M_{d^2}.$$

The **jointly completely bounded norm** is

$$\|u\|_{\text{jcb}} = \sup_{d \geq 1} \|u_d\|.$$

Equivalently, $\|u\|_{\text{jcb}}$ is the completely bounded norm of the associated map $E \rightarrow F^*$, where F^* carries its canonical operator-space structure.

The finite-dimensional operator-space Grothendieck theorem of Pisier and Shlyakhtenko says that there is a universal constant C such that every jointly completely bounded bilinear form $u : A \times B \rightarrow \mathbb{C}$ on finite-dimensional C^* -algebras admits states f_1, f_2 on A and g_1, g_2 on B satisfying [22]

$$|u(a, b)| \leq C \|u\|_{\text{jcb}} (f_1(a^* a) + f_2(a a^*))^{1/2} (g_1(b^* b) + g_2(b b^*))^{1/2}.$$

This is the noncommutative analogue of the classical Grothendieck inequality, but the Bell scenarios governed by the two theories differ:

$$\begin{array}{ll} \text{binary correlations} & \longleftrightarrow \text{classical Grothendieck inequality,} \\ \text{multi-output correlations} & \longleftrightarrow \text{operator-space norms.} \end{array}$$

Unlike the binary correlation setting, general multi-output Bell inequalities need not have a dimension-independent quantum-to-classical ratio [23].

2.9 Von Neumann algebras and subfactors

For $\mathcal{S} \subseteq \mathcal{B}(H)$, its **commutant** is

$$\mathcal{S}' = \{T \in \mathcal{B}(H) : Tx = xT \text{ for every } x \in \mathcal{S}\}.$$

A **von Neumann algebra** is a unital $*$ -subalgebra $M \subseteq \mathcal{B}(H)$ satisfying $M = M''$ [12]. A net (T_λ) converges to T in the **strong operator topology** if $T_\lambda \xi \rightarrow T\xi$ for every $\xi \in H$, and in the **weak operator topology** if

$$\langle T_\lambda \xi, \eta \rangle \rightarrow \langle T\xi, \eta \rangle$$

for every $\xi, \eta \in H$.

Exercise 23 (Bicommutant theorem)

Let $A \subseteq \mathcal{B}(H)$ be a unital $*$ -subalgebra. Prove that

$$A'' = \overline{A}^{\text{SOT}} = \overline{A}^{\text{WOT}}.$$

The **center** of a von Neumann algebra is $Z(M) = M \cap M'$, and M is a **factor** if $Z(M) = \mathbb{C}1$. A positive functional or positive map is **normal** if it preserves suprema of bounded increasing nets of positive elements, and a positive functional ψ is **faithful** if $\psi(a^*a) = 0$ implies $a = 0$. A **type II₁ factor** is an infinite-dimensional factor with a faithful normal tracial state

$$\tau : M \rightarrow \mathbb{C}, \quad \tau(ab) = \tau(ba), \quad \tau(1) = 1.$$

The normalized trace is an analytic analogue of the categorical trace in Algebra, not an identification with it.

For projections $p, q \in M$, write $p \sim q$ if there is $v \in M$ such that

$$v^*v = p, \quad vv^* = q.$$

Write $p \precsim q$ if p is equivalent to a projection r satisfying $rq = r$.

Exercise 24 (Projection dimension)

Let M be a type II₁ factor with normalized trace τ . Prove that

$$p \sim q \implies \tau(p) = \tau(q),$$

and prove the comparison theorem

$$p \precsim q \iff \tau(p) \leq \tau(q).$$

Deduce that $\tau(p) = \tau(q)$ if and only if $p \sim q$.

A von Neumann algebra is **finite** if $v^*v = 1$ implies $vv^* = 1$. Let $N \subseteq M$ be finite von Neumann algebras with a common normalized trace τ . A **trace-preserving conditional expectation** is a normal unital completely positive map $E_N : M \rightarrow N$ such that

$$\tau \circ E_N = \tau, \quad E_N(n_1 x n_2) = n_1 E_N(x) n_2$$

for $x \in M$ and $n_1, n_2 \in N$. Let $L^2(M, \tau)$ be the completion of M for

$$\langle x, y \rangle_\tau = \tau(y^*x).$$

Let M act on $L^2(M, \tau)$ by left multiplication.

Exercise 25 *Prove that E_N extends to the orthogonal projection*

$$e_N : L^2(M, \tau) \longrightarrow L^2(N, \tau).$$

Prove that

$$e_N = e_N^* = e_N^2, \quad e_N x e_N = E_N(x) e_N$$

for every $x \in M$.

The **basic construction** is

$$M_1 = \langle M, e_N \rangle'' \subseteq \mathcal{B}(L^2(M, \tau)),$$

the double commutant of the algebra generated by M and e_N . For a projection $p = [p_{ij}] \in M_k(N)$ acting by left multiplication, define

$$\dim_N(pL^2(N, \tau)^k) = \sum_{i=1}^k \tau(p_{ii}).$$

An inclusion $N \subseteq M$ has **finite Jones index** if $L^2(M, \tau)$ is isomorphic as a right N -module to such a module; then

$$[M : N] = \dim_N L^2(M, \tau).$$

Exercise 26 (Basic construction)

Let $N \subseteq M$ be an inclusion of type II_1 factors of finite Jones index. Prove that M_1 is a finite factor and that its normalized trace satisfies [13]

$$\tau_1(xe_N y) = \frac{1}{[M : N]} \tau(xy)$$

for $x, y \in M$.

Iterating the basic construction gives the **Jones tower**

$$N = M_{-1} \subset M = M_0 \subset M_1 \subset M_2 \subset \cdots.$$

Let e_i be the Jones projection used to construct M_i from M_{i-1} , and set

$$\delta = \sqrt{[M : N]}.$$

The **Temperley–Lieb algebra** with loop parameter δ is generated by U_i subject to

$$U_i^2 = \delta U_i, \quad U_i U_j = U_j U_i \quad (|i - j| \geq 2), \quad U_i U_{i \pm 1} U_i = U_i.$$

Exercise 27 (Temperley–Lieb relations)

Prove that the Jones projections satisfy [16, 14]

$$e_i^2 = e_i = e_i^*, \quad e_i e_j = e_j e_i \quad (|i - j| \geq 2),$$

$$e_i e_{i \pm 1} e_i = \delta^{-2} e_i.$$

Deduce that $U_i \mapsto \delta e_i$ defines a representation of the Temperley–Lieb algebra.

Let $q \in \mathbb{C}^\times$ satisfy $\delta = q + q^{-1}$ and define

$$\sigma_i = q 1 - U_i.$$

These operators reconnect the Temperley–Lieb construction with the braid groups introduced in Algebra.

Exercise 28 (Braid representation)

Prove that σ_i is invertible with inverse $q^{-1} 1 - U_i$, and prove that

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1},$$

$$\sigma_i \sigma_j = \sigma_j \sigma_i \quad (|i - j| \geq 2).$$

Deduce that the σ_i define a representation of the braid group [15].

The compatible normalized traces on the Jones tower restrict to a positive **Markov trace** on the Temperley–Lieb algebras satisfying

$$\text{tr}(x U_i) = \delta^{-1} \text{tr}(x)$$

whenever x belongs to the algebra generated by the preceding generators. Define

$$P_0(\delta) = 1, \quad P_1(\delta) = \delta, \quad P_{n+1}(\delta) = \delta P_n(\delta) - P_{n-1}(\delta).$$

Exercise 29 Prove that, when $\sin \theta \neq 0$,

$$P_n(2 \cos \theta) = \frac{\sin((n+1)\theta)}{\sin \theta}.$$

Use positivity of the Markov trace to prove that if $0 < \delta < 2$, then

$$\delta = 2 \cos \left(\frac{\pi}{m} \right)$$

for some integer $m \geq 3$.

Exercise 30 (Jones index theorem)

Let $N \subseteq M$ be an inclusion of type II_1 factors of finite Jones index. Prove that [14, 13]

$$[M : N] \in \left\{ 4 \cos^2 \left(\frac{\pi}{n} \right) : n \geq 3 \right\} \cup [4, \infty).$$

For

$$[M : N] = 4 \cos^2 \left(\frac{\pi}{k+2} \right), \quad \delta = \sqrt{[M : N]},$$

prove that the Jones projections determine a representation of the Temperley–Lieb algebra $\text{TL}_n(\delta)$, and hence a braid-group representation. Show that the Markov trace of a braid closure satisfies the Jones skein relation. Relate the parameter

$$\delta = 2 \cos \left(\frac{\pi}{k+2} \right)$$

to the $SU(2)_k$ fusion and braiding data from Algebra and to the Wilson-loop invariant of Chern–Simons theory from Geometry.

3 Geometry

A **smooth manifold** is a topological space that is locally diffeomorphic to Euclidean space $\mathbb{R}^n$. A **smooth vector bundle** $p : E \rightarrow M$ consists of a smooth surjection p between smooth manifolds E (the total space) and M (the base space) such that all fibers are isomorphic to a single vector space V , and smooth local trivializations exist (locally, the fiber bundle looks like a product space). For a smooth map $f : N \rightarrow M$, the **pullback bundle** is

$$f^*E = \{(x, e) \in N \times E : f(x) = p(e)\} \longrightarrow N.$$

A prototypical example is the **tangent bundle** TM which consists of all tangent spaces of points of a given manifold. The **cotangent bundle** T^*M is the dual bundle of TM . A **vector field** $X \in \Gamma(TM)$ is a smooth section of the tangent bundle (i.e. a smooth map $s : M \rightarrow TM$ such that $p \circ s = 1_M$) [3]. An orientation of M is a smoothly varying choice of orientation of each tangent space, using the determinant lines defined in the Algebra section.

Exercise 31 Write down precise definitions for the following:

- smooth manifold, smooth map, vector bundle, tangent bundle.
- bundle map, pullback, direct sum, tensor product, and dual bundle
- derivative of a smooth function between manifolds

For a compact Hausdorff space X , let $\text{Vect}(X)$ be the category of finite-rank complex vector bundles with continuous local trivializations over X . It has direct sums, tensor products, duals, and pullbacks, and is an additive symmetric monoidal category with duals [7]. The **Grothendieck group** $K^0(X)$ is generated by isomorphism classes $[E]$ subject to

$$[E \oplus F] = [E] + [F].$$

Tensor product defines $[E][F] = [E \otimes F]$, making $K^0(X)$ a commutative ring. For a continuous map $f : X \rightarrow Y$ between compact Hausdorff spaces, pullback defines

$$f^* : K^0(Y) \longrightarrow K^0(X), \quad [E] \longmapsto [f^* E].$$

Exercise 32 (Grothendieck group of vector bundles)

Let X and Y be compact Hausdorff spaces and let $f : X \rightarrow Y$ be continuous. Prove that direct sum and tensor product induce a commutative ring structure on $K^0(X)$ and that f^* is a ring homomorphism. Prove that

$$K^0(\text{pt}) \cong \mathbb{Z}$$

via the rank map.

Fix a basepoint $x_0 \in X$, regarded as a map $x_0 : \text{pt} \rightarrow X$. The **reduced K-theory** of the pointed space (X, x_0) is

$$\tilde{K}^0(X) = \ker \left(K^0(X) \xrightarrow{x_0^*} K^0(\text{pt}) \right).$$

For a bundle E of rank r , its **reduced class** is

$$[E] - [\mathbb{C}^r],$$

where $\mathbb{C}^r = X \times \mathbb{C}^r$ is the trivial rank- r bundle. Topological K-theory extends K^0 to generalized cohomology groups $K^n(X)$. Bott periodicity gives

$$K^{n+2}(X) \cong K^n(X),$$

so complex K-theory is naturally $\mathbb{Z}/2$ -graded.

Exercise 33 (Stable equivalence)

Let E and F be complex vector bundles over a compact Hausdorff space X . Prove that $[E] = [F]$ in $K^0(X)$ if and only if there is a vector bundle G such that

$$E \oplus G \cong F \oplus G.$$

Deduce that the reduced class of E is unchanged after replacing E by $E \oplus \mathbb{C}^n$.

For the gapped families below, assume that X is connected, so vector-bundle ranks are constant. Let V be a finite-dimensional Hermitian vector space. A continuous family is a map

$$H : X \longrightarrow \text{End}(V)$$

and we write $H_x = H(x)$. It is **gapped** if every H_x is self-adjoint and invertible. Compactness gives numbers $\delta, M > 0$ such that every eigenvalue of every H_x lies in $[-M, -\delta] \cup [\delta, M]$. Choose a positively oriented contour Γ enclosing $[-M, -\delta]$ and excluding $[\delta, M]$, and define

$$P_x^- = \frac{1}{2\pi i} \int_{\Gamma} (z - H_x)^{-1} dz, \quad E_x^- = \text{im}(P_x^-).$$

Exercise 34 (Negative spectral bundle)

Let H be a gapped family. Prove that

$$(P_x^-)^2 = P_x^-, \quad (P_x^-)^* = P_x^-,$$

that $x \mapsto P_x^-$ is continuous, and that

$$E_H^- = \coprod_{x \in X} E_x^-$$

is a complex vector bundle over X . Prove that its rank is locally constant.

For a gapped family $H : X \times [0, 1] \rightarrow \text{End}(V)$, let E_j^- be the negative spectral bundle of $x \mapsto H(x, j)$, and let $i_j : X \rightarrow X \times [0, 1]$ be given by $i_j(x) = (x, j)$.

Exercise 35 (Gap-preserving homotopy)

Prove that the negative spectral subspaces of H form a vector bundle $E^- \rightarrow X \times [0, 1]$. Prove that the endpoint pullbacks of any vector bundle over $X \times [0, 1]$ are isomorphic, and apply this to $i_0^* E^-$ and $i_1^* E^-$. Deduce that

$$[E_0^-] = [E_1^-] \in K^0(X).$$

Let $Q \in \text{End}(W)$ be a fixed invertible self-adjoint endomorphism of a finite-dimensional Hermitian space W , let W^- be its negative eigenspace, and write $\underline{W}^- = X \times W^-$.

Exercise 36 (Stabilization)

For a gapped family H , prove that

$$E_{H \oplus Q}^- \cong E_H^- \oplus \underline{W}^-.$$

Deduce that

$$[E_H^-] - [\mathbb{C}^{\text{rank } E_H^-}] \in \tilde{K}^0(X)$$

is unchanged by adjoining a constant gapped summand.

Let $E \subseteq X \times V$ be a subbundle and let P_x be the orthogonal projection onto E_x .

Exercise 37 (Flattened family)

Define $H_x = 1_V - 2P_x$. Prove that $x \mapsto H_x$ is a gapped family, that

$$H_x^2 = 1_V,$$

and that its negative spectral bundle is E . Deduce that every subbundle of a trivial bundle arises as the negative spectral bundle of a gapped family.

Thus a finite-dimensional gapped family determines a class in $\tilde{K}^0(X)$ that is invariant under gap-preserving homotopy and stabilization. Conversely, every subbundle of a trivial bundle occurs as a negative spectral bundle.

In a finite-dimensional gapped family, the negative spectral subspaces form a vector bundle. Gap-preserving deformation preserves its stable isomorphism class, so the associated reduced K-theory class is a topological invariant. This construction underlies the classification of noninteracting topological insulators and superconductors under specified symmetry assumptions [8].

The exterior powers $\Lambda^k T_p^* M$ are those defined in the Algebra section. To define integration in a coordinate-free way, we will integrate differential forms over chains. The space of **differential n -forms** is $\Omega^n(M) = \Gamma(\Lambda^n T^* M)$. On an oriented n -manifold, a **volume form** is a nowhere-vanishing element of $\Omega^n(M)$ that is positive with respect to the orientation. The exterior derivative $d : \Omega^n(M) \rightarrow \Omega^{n+1}(M)$ extends the differential of smooth functions and satisfies [6]

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta$$

for $\alpha \in \Omega^p(M)$. A **singular n -simplex** is a smooth map $\sigma : \Delta^n \rightarrow M$ where Δ^n is the standard n -simplex. The collection of singular n -simplices in M freely generates the abelian group $C_n(M)$ of **singular n -chains** in M . The **boundary map** $\partial : C_n(M) \rightarrow C_{n-1}(M)$ is defined on each simplex $\sigma : \Delta^n \rightarrow M$ by

$$\partial\sigma = \sum_{j=0}^n (-1)^j \sigma \circ f_j,$$

where $f_j : \Delta^{n-1} \rightarrow \Delta^n$ is the inclusion of the face opposite the j -th vertex of Δ^n .

A **chain complex** $(A_n, d_n)_n$ is a sequence of abelian groups A_n and homomorphisms $d_n : A_n \rightarrow A_{n-1}$ such that $d_{n-1} \circ d_n = 0$ for all n . The n -th **homology group** H_n of the chain complex is

$$H_n = \ker(d_n) / \operatorname{im}(d_{n+1}).$$

The prototypical example is singular homology which corresponds to the chain complex of singular chains with the boundary map defined above.

Cohomology and cochains are defined similarly, using superscripts instead of subscripts and boundary maps $d^n : A^n \rightarrow A^{n+1}$ increase degree instead of decrease it. Singular cohomology corresponds to the cochain complex given by dualizing the singular chain complex above.

The **de Rham cohomology** of M is

$$H_{\text{dR}}^n(M) = \ker(d : \Omega^n(M) \rightarrow \Omega^{n+1}(M)) / \operatorname{im}(d : \Omega^{n-1}(M) \rightarrow \Omega^n(M)).$$

Exercise 38 *Prove the following:*

- Show that $d^2 = 0$, so differential forms on a manifold M form a cochain complex.

- Define the integral of an n -form on an n -chain. Prove that $\int_c d\omega = \int_{\partial c} \omega$ for any n -chain c and differential $n-1$ -form ω .
- Singular cohomology with real coefficients and de Rham cohomology are naturally isomorphic.
- Bounded chain complexes of finite-dimensional vector spaces form a symmetric monoidal abelian category with duals, the trace of an endomorphism is its Lefschetz number, and the dimension of a chain complex is its Euler characteristic.

For a vector field X , contraction is the map $\iota_X : \Omega^p(M) \rightarrow \Omega^{p-1}(M)$ defined by

$$(\iota_X \alpha)(X_2, \dots, X_p) = \alpha(X, X_2, \dots, X_p).$$

The Lie derivative of forms is determined by Cartan's formula $\mathcal{L}_X = d\iota_X + \iota_X d$.

A **symplectic form** is a two-form $\omega \in \Omega^2(M)$ such that $d\omega = 0$ and $X \mapsto \iota_X \omega$ is a bundle isomorphism $TM \rightarrow T^*M$. Write $C^\infty(M)$ for the smooth real-valued functions on M . For $H \in C^\infty(M)$, define X_H by

$$\iota_{X_H} \omega = dH.$$

Exercise 39 (Liouville's theorem)

Let (M, ω) be $2n$ -dimensional with ω symplectic. Prove that $\omega^n/n!$ is a volume form. For $H \in C^\infty(M)$, prove using Cartan's formula that [26]

$$\mathcal{L}_{X_H} \omega = 0.$$

Deduce that $\mathcal{L}_{X_H} \omega^n = 0$ and hence that the local flow of X_H preserves the volume form

$$\frac{\omega^n}{n!}.$$

For $f, g \in C^\infty(M)$, define

$$\{f, g\} = \omega(X_f, X_g).$$

The following is a Hamiltonian symmetry-conservation statement, not the full variational theorem of [27].

Exercise 40 (Hamiltonian Noether theorem)

Let (M, ω) be symplectic. Prove that [26]

$$X_g(f) = \{f, g\}.$$

Let Φ_t be the local flow of X_f . Prove that the following are equivalent:

- $H \circ \Phi_t = H$ whenever both sides are defined.
- $X_f(H) = 0$.

- $\{H, f\} = 0$.
- f is constant along the integral curves of X_H .

A **pseudo-Riemannian metric** is a smooth nondegenerate symmetric section $g \in \Gamma(T^*M \otimes T^*M)$. On an oriented pseudo-Riemannian n -manifold, the metric induces a bilinear form $\langle \cdot, \cdot \rangle_g$ on differential forms. The **metric volume form** vol_g is the unique positive volume form satisfying

$$|\langle \text{vol}_g, \text{vol}_g \rangle_g| = 1.$$

The Hodge star $\star$ is determined by

$$\alpha \wedge \star \beta = \langle \alpha, \beta \rangle_g \text{vol}_g.$$

Exercise 41 (Charge conservation from Maxwell's equations)

Let M be an oriented four-dimensional pseudo-Riemannian manifold. Suppose that $F \in \Omega^2(M)$ and $J \in \Omega^3(M)$ satisfy [28]

$$dF = 0, \quad d \star F = J.$$

Prove that $dJ = 0$. If C is an oriented compact four-dimensional chain, deduce that

$$\int_{\partial C} J = 0.$$

If the three-dimensional pieces of its boundary satisfy

$$\partial C = \Sigma_1 - \Sigma_0 + B,$$

prove that

$$\int_{\Sigma_1} J - \int_{\Sigma_0} J = - \int_B J.$$

For a closed form A , write $[A]$ for its de Rham cohomology class. The next exercise is a modern de Rham formulation; the original paper does not use this language [36, 28].

Exercise 42 (Aharonov–Bohm cohomological obstruction) Let M be a compact smooth manifold and let $A \in \Omega^1(M)$ satisfy $dA = 0$. Prove the following:

- For every contractible open set $U \subseteq M$, there exists $\lambda \in C^\infty(U)$ such that $A|_U = d\lambda$.
- For every smooth closed curve γ , the number $\int_\gamma A$ depends only on the homology class of γ .
- There exists a global $\lambda \in C^\infty(M)$ with $A = d\lambda$ iff $[A] = 0 \in H_{\text{dR}}^1(M)$.
- If $[A] \neq 0$, then there exists a smooth closed curve γ such that $\int_\gamma A \neq 0$.

An **affine connection** is a map $\nabla : \Gamma(TM) \times \Gamma(TM) \rightarrow \Gamma(TM)$ such that

$$\nabla_{fX}Y = f\nabla_XY, \quad \nabla_X(fY) = X(f)Y + f\nabla_XY.$$

For a covector field α , set

$$(\nabla_X\alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_XY).$$

The connection extends uniquely to all tensor fields by the product rule and by commuting with contractions; this defines the **covariant derivative** of a tensor field.

The **Levi-Civita connection** of a pseudo-Riemannian metric is the unique affine connection satisfying

$$\nabla_XY - \nabla_YX = [X, Y], \quad \nabla g = 0.$$

An affinely parametrized curve $\gamma : I \rightarrow M$ is a **geodesic** if $\nabla_{\dot{\gamma}}\dot{\gamma} = 0$. The Lie derivative of the metric is

$$(\mathcal{L}_Xg)(Y, Z) = X(g(Y, Z)) - g([X, Y], Z) - g(Y, [X, Z]),$$

and X is a **Killing vector field** if $\mathcal{L}_Xg = 0$.

Let (M, g) be a pseudo-Riemannian manifold with Levi-Civita connection ∇ . Curvature governs the relative acceleration of geodesics, while symmetries of g determine quantities preserved along them [29, 30].

Exercise 43 (Killing fields and conserved quantities)

Let X be a Killing vector field and let γ be an affinely parametrized geodesic. Prove that

$$\frac{d}{dt}g(\dot{\gamma}, \dot{\gamma}) = 0, \quad \frac{d}{dt}g(X, \dot{\gamma}) = 0.$$

Deduce that $g(X, \dot{\gamma})$ is constant along γ .

The **curvature** of ∇ is the map

$$R(X, Y)Z = \nabla_X\nabla_YZ - \nabla_Y\nabla_XZ - \nabla_{[X, Y]}Z.$$

Exercise 44 (Curvature as a tensor)

For $f \in C^\infty(M)$ and vector fields X, Y, Z , prove that

$$R(fX, Y)Z = fR(X, Y)Z, \quad R(X, fY)Z = fR(X, Y)Z,$$

$$R(X, Y)(fZ) = fR(X, Y)Z.$$

Verify explicitly that every derivative of f cancels. Prove that $R(X, Y) = -R(Y, X)$ and deduce that

$$R \in \Gamma(\Lambda^2 T^*M \otimes T^*M \otimes TM).$$

Let $\Gamma : (-\varepsilon, \varepsilon) \times I \rightarrow M$ be a smooth variation through affinely parametrized geodesics, and define the vector fields along Γ by

$$T = \frac{\partial \Gamma}{\partial t}, \quad J = \frac{\partial \Gamma}{\partial s}.$$

Then $[T, J] = 0$.

Exercise 45 (Geodesic deviation) *For a variation Γ as above, prove that*

$$\nabla_T \nabla_T J = R(T, J)T.$$

Thus curvature determines the relative acceleration of nearby geodesics.

The trace used below is the ordinary trace defined in the Algebra section. Define the **Ricci tensor**, **scalar curvature**, and **Einstein tensor** by

$$\text{Ric}(X, Y) = \text{tr}(Z \mapsto R(Z, X)Y), \quad S = \text{tr}_g(\text{Ric}), \quad G = \text{Ric} - \frac{1}{2}Sg.$$

For a symmetric covariant two-tensor A , its **divergence** is the covector field

$$(\text{div } A)(Y) = \text{tr}_g((X, Z) \mapsto (\nabla_X A)(Z, Y)).$$

Exercise 46 (Contracted Bianchi identity and conservation)

Assume the second Bianchi identity

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0.$$

Contract this identity to prove that

$$\text{div } G = 0.$$

Deduce that if T is a symmetric covariant two-tensor satisfying

$$G + \Lambda g = \kappa T$$

for constants Λ, κ with $\kappa \neq 0$, then $\text{div } T = 0$.

Connections and curvature also determine characteristic classes. In odd dimensions, their transgression forms define Chern–Simons functionals.

Let $E \rightarrow M$ be a complex vector bundle and write $\Omega^0(M, E) = \Gamma(E)$. A **connection** on E is a $\mathbb{C}$ -linear map

$$\nabla : \Omega^0(M, E) \longrightarrow \Omega^1(M, E)$$

satisfying $\nabla(fs) = df \otimes s + f\nabla s$. For $\alpha \in \Omega^p(M)$ and an E -valued form η , extend it by

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^p \alpha \wedge \nabla \eta.$$

Its **curvature** is

$$F_\nabla = \nabla^2 \in \Omega^2(M, \text{End } E).$$

The connection induces one on $\text{End } E$ -valued forms.

Exercise 47 (Bianchi identity) Let ∇ be a connection on $E \rightarrow M$. Prove that

$$\nabla F_\nabla = 0.$$

Let P be an invariant symmetric k -linear polynomial on the endomorphism algebra of a fiber of E , where invariant means invariant under simultaneous conjugation. Define [31]

$$P(F_\nabla) = P(F_\nabla, \dots, F_\nabla) \in \Omega^{2k}(M).$$

Exercise 48 (Chern–Weil theorem)

Prove that

$$dP(F_\nabla) = 0.$$

For two connections ∇^0, ∇^1 on E , prove that

$$[P(F_{\nabla^0})] = [P(F_{\nabla^1})] \in H_{\text{dR}}^{2k}(M).$$

Define the **Chern character form** by

$$\text{ch}(E, \nabla) = \text{tr} \exp \left(\frac{iF_\nabla}{2\pi} \right).$$

When induced connections are understood, suppress them from the notation.

Exercise 49 (Chern character)

Prove that $d \text{ch}(E, \nabla) = 0$ and that its de Rham class is independent of ∇ . For a second complex vector bundle F with connection, prove that

$$\text{ch}(E \oplus F) = \text{ch}(E) + \text{ch}(F),$$

$$\text{ch}(E \otimes F) = \text{ch}(E) \wedge \text{ch}(F).$$

If M is compact, deduce that the Chern character defines a ring homomorphism

$$\text{ch} : K^0(M) \longrightarrow H_{\text{dR}}^{\text{even}}(M), \quad H_{\text{dR}}^{\text{even}}(M) = \bigoplus_{j \geq 0} H_{\text{dR}}^{2j}(M).$$

For two connections on E , set

$$\nabla^t = (1-t)\nabla^0 + t\nabla^1, \quad A = \nabla^1 - \nabla^0, \quad F_t = F_{\nabla^t},$$

where $A \in \Omega^1(M, \text{End } E)$, and define

$$\text{CS}_P(\nabla^0, \nabla^1) = k \int_0^1 P(A, F_t, \dots, F_t) dt.$$

Thus $\text{CS}_P(\nabla^0, \nabla^1) \in \Omega^{2k-1}(M)$.

Exercise 50 (Chern–Simons transgression)

Prove that

$$P(F_{\nabla^1}) - P(F_{\nabla^0}) = d \text{CS}_P(\nabla^0, \nabla^1).$$

Let $\mathfrak{g} \subseteq M_N(\mathbb{C})$ be a matrix Lie algebra whose trace pairing is nondegenerate. For $A \in \Omega^1(M, \mathfrak{g})$, wedge products include matrix multiplication. Define

$$F_A = dA + A \wedge A$$

and

$$\text{CS}(A) = \text{tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right).$$

Exercise 51 (Chern–Simons form)

Prove that

$$d \text{CS}(A) = \text{tr}(F_A \wedge F_A).$$

For a closed oriented three-manifold M , define

$$S_{\text{CS}}(A) = \int_M \text{CS}(A).$$

Exercise 52 (Chern–Simons equation)

For $\alpha \in \Omega^1(M, \mathfrak{g})$, set $A_t = A + t\alpha$. Prove that

$$\left. \frac{d}{dt} \right|_{t=0} S_{\text{CS}}(A_t) = 2 \int_M \text{tr}(\alpha \wedge F_A).$$

Deduce that the critical points satisfy $F_A = 0$.

Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$. For a smooth map $g : M \rightarrow G$, define

$$A^g = g^{-1}Ag + g^{-1}dg.$$

Assume that G is compact, connected, simple, and simply connected, and normalize the invariant trace so that

$$\nu(g) = \frac{1}{24\pi^2} \int_M \text{tr}((g^{-1}dg)^3) \in \mathbb{Z}.$$

For $k \in \mathbb{R}$, set $S_k = (k/4\pi)S_{\text{CS}}$.

Exercise 53 (Gauge transformation)

Prove that

$$\text{CS}(A^g) - \text{CS}(A) = -d \text{tr}(dg g^{-1} \wedge A) - \frac{1}{3} \text{tr}((g^{-1}dg)^3).$$

Deduce that $\exp(iS_k)$ is invariant under all gauge transformations when $k \in \mathbb{Z}$.

Now restrict to $2 + 1$ spacetime dimensions. Let M be a closed oriented three-manifold and let $\ell > 0$, so the cosmological constant is $\Lambda = -\ell^{-2}$. After identifying the coframe representation with $\mathfrak{sl}(2, \mathbb{R})$, let

$$e, \omega \in \Omega^1(M, \mathfrak{sl}(2, \mathbb{R})), \quad A^\pm = \omega \pm \frac{1}{\ell} e.$$

The pair (A^+, A^-) has gauge Lie algebra $\mathfrak{sl}(2, \mathbb{R}) \oplus \mathfrak{sl}(2, \mathbb{R})$. Write

$$d_\omega e = de + \omega \wedge e + e \wedge \omega, \quad F_\omega = d\omega + \omega \wedge \omega.$$

Exercise 54 (Three-dimensional gravity)

Prove that $F_{A^+} = F_{A^-} = 0$ is equivalent to

$$d_\omega e = 0, \quad F_\omega + \frac{1}{\ell^2} e \wedge e = 0.$$

Assuming that $e : TM \rightarrow \mathfrak{sl}(2, \mathbb{R})$ is fiberwise invertible, identify these equations with the torsion-free condition and the Einstein equation with constant negative curvature.

Define the first-order three-dimensional Einstein–Hilbert functional, up to its conventional overall coefficient, by

$$S_{\text{EH}}(e, \omega) = \int_M \text{tr} \left(e \wedge F_\omega + \frac{1}{3\ell^2} e \wedge e \wedge e \right).$$

Exercise 55 (Gravity as Chern–Simons theory)

Prove that, up to an overall normalization and an exact term,

$$S_{\text{CS}}(A^+) - S_{\text{CS}}(A^-)$$

equals $S_{\text{EH}}(e, \omega)$ [34, 35].

This is a local Lie-algebra formulation of gravity in three spacetime dimensions. Changing between Lorentzian and Euclidean signatures changes the relevant real form and reality conditions. It is an analytic continuation, not a statement that four-dimensional gravity is obtained from Chern–Simons theory by Wick rotation.

For a connection A and a finite-dimensional representation V of its gauge group, parallel transport around an oriented closed curve γ gives the holonomy $\text{Hol}_\gamma(A)$. Define the **Wilson loop**

$$W_V(\gamma, A) = \text{tr}_V(\text{Hol}_\gamma(A)).$$

For a framed oriented link $L = \gamma_1 \sqcup \cdots \sqcup \gamma_r$ labeled by $V_1, \dots, V_r$, set

$$W_L(A) = \prod_{j=1}^r W_{V_j}(\gamma_j, A).$$

At the level of this document, the Chern–Simons expectation value

$$\langle W_L \rangle_k = \frac{\int e^{iS_k(A)} W_L(A) \mathcal{D}A}{\int e^{iS_k(A)} \mathcal{D}A}$$

is a formal path-integral expression, and its value depends on the chosen framing [32]. For normalized $SU(2)$ theory at integer level $k \geq 1$, set

$$q = \exp \left(\frac{2\pi i}{k+2} \right).$$

Exercise 56 (Wilson loops and the Jones polynomial)

Let L_+ , L_- , and L_0 differ only by a positive crossing, a negative crossing, and the oriented smoothing. Assume that, after the standard framing normalization, the Wilson-loop expectation values satisfy

$$q^{-1}\langle W_{L_+} \rangle_k - q\langle W_{L_-} \rangle_k = (q^{1/2} - q^{-1/2}) \langle W_{L_0} \rangle_k$$

and $\langle W_{\bigcirc} \rangle_k = 1$. Prove that these relations determine the Jones polynomial of an oriented link.

Chern–Simons theory relates the three parts of the document. Its Wilson lines carry braid-group actions governed algebraically by braided tensor categories, while the resulting link invariants are realized analytically through Temperley–Lieb algebras and subfactors [33]. These connections are indicated here, not proved as a full equivalence.

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